

Day-ahead Load Forecasting using Explainable Artificial Intelligence

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Abstract—A real-time balance between power production and consumption is needed to ensure stable power flow and frequency. Due to the fact that electricity is needed to be consumed at a very instant time, accurate energy forecasting is required. Meanwhile, the recent high integration of Renewable Energy Sources (RES) into electrical grids introduces the accuracy problem in forecasts due to their inherited volatile nature. To overcome these challenges, digitalization of the energy sector started with implementation of emerging technologies such as Artificial Intelligence (AI), Information and Communication Technology (ICT), and the Internet of Things (IoT) for a better forecasting. However, the perceived black-box nature of AI gave rise to increased demand on understanding the inner working mechanics of them. This paper presents a use case of day-ahead load forecasting using an eXplainable Artificial Intelligence (XAI) tool, i.e., SHapley Additive exPlanations (SHAP), which helps understand the opaque, complex nature of the AI system by interpreting the feature importance.

Index Terms—Energy forecasting, Explainable Artificial Intelligence (XAI), explainability and transparency, feature importance, load forecasting

I. INTRODUCTION

Grid stability is essential for stable operation of the power system. This is achieved through real-time balance of power supply and load demand governed by the principles of the free-market and by Transmission System Operators (TSO). However, due to the recent changes in the energy industry, such as increased penetration of Distributed Energy Resources (DER) which encompass volatile energy technologies such as solar photovoltaics (PV) and wind, the challenge of balancing supply and demand of power is increasing. Therefore, an accurate forecasting of both generation and demand side is necessary to ensure stability of the power grid.

To overcome these challenges, digitalization of the energy sector started with the implementation and integration of emerging digital technologies such as Internet of Things (IoT), Information and Communication Technologies (ICT) and Artificial Intelligence (AI). The initial applications of AI

were focused on statistical methods, hence also interpretable to humans. However, recent advances in black box AI, such as deep neural networks, give rise to increased demand in understanding the underlying nature of the models as forecasting models keep evolving into more sophisticated versions [1].

As mentioned, AI methods are able to analyze, predict and forecast both load and power generation with high accuracy [2]. However, most methods are still sourced from the black box nature of AI and cannot provide satisfying levels of interpretability and transparency [3]. By using eXplainable Artificial Intelligence (XAI) tools, a light can be shed on the inner workings of the system and provide knowledge of the model, which can be used to further improve the performance and stability of load and power generation forecasts.

In the literature, AI methods have been extensively used by researchers for load and energy generation forecasts. Reference [4] presents a short-term load forecasting model, i.e., between one day to one week ahead, using Least Square Support Vector Machine (LS-SVM) combined with fuzzy control. They found that the combination has the potential to provide better accuracy than a single model. Meanwhile, [5] proposes another short-term load forecasting model by combining Artificial Neural Networks (ANN) with SVM, which provides better results both in speed and forecasting accuracy than individual methods. The authors in [6] propose a weekly peak load forecasting model by using deep learning algorithms and compared the algorithms performance the results indicate the proposed forecasting method provided a better forecasting performance than the multiple regression method. Researchers in [7] investigated an hour-ahead PV power generation forecasting model using several methods such as SVR, polynomial regression and Lasso.

Although AI techniques have been extensively implemented and researched, there are still limited studies using XAI methods in the fields of load and energy forecasting. The authors in [8], [9] investigate several XAI methods (SHAP, LIME,

and ELI5) in order to interpret solar PV power forecasting. Further, [10] propose a load forecasting model using eXtreme Gradient Boosting (XGBoost) and interpret it using the XAI framework SHapley Additive exPlanations (SHAP).

This paper presents a day-ahead load forecasting model analyzed using an XAI tool, SHAP. The paper aims to facilitate deployment of AI-based forecasting with the use of XAI methods. The main contributions of this paper include: (1) The development of a day-ahead load forecasting model; (2) applying an XAI method to a load forecasting model; (3) evaluation of the model using all features and a subset of the features; and (4) demonstrating the proposed model on a Norwegian case study of one price region (NO1), and investigating the impact of Easter week on the model by utilizing XAI.

II. MACHINE LEARNING MODELS AND EXPLAINABLE ARTIFICIAL INTELLIGENCE

A. Multilayer perceptron

A widely used type of neural network is the MultiLayer Perceptron (MLP). The MLP consists of input, output and a number of other "hidden" layers between the two. Each hidden layer has a distinct number of units determining the number of outputs of that layer. For each unit, a weighted linear combinations of its inputs is made, where the result is fed through a non-linear activation function before being passed to the subsequent layer. This process is repeated until the output layer where a linear (for regression problems) activation function is used instead. In a fully connected network, there are connections between all units of two subsequent layers. In vectorized form, the set of equations describing this type of network may be written as

$$\mathbf{h}^{[1]} = \phi^{[1]}(\mathbf{W}^{[1]}\mathbf{x} + \mathbf{b}^{[1]}) \quad (1)$$

$$\mathbf{h}^{[l]} = \phi^{[l]}(\mathbf{W}^{[l]}\mathbf{h}^{[l-1]} + \mathbf{b}^{[l]}) \quad (2)$$

$$\mathbf{y} = \phi^{[L]}(\mathbf{W}^{[L]}\mathbf{h}^{[L-1]} + \mathbf{b}^{[L]}) \quad (3)$$

where L is the number of layers in the network; $\mathbf{h}^{[l]}$ is the output vector of layer l ; $\mathbf{W}$ denotes the weight matrix of a layer, where w_{ij} is the weight connecting unit j of the preceding layer to unit i ; $\mathbf{b}$ is the bias vector of a layer; and ϕ is the element-wise activation function [11].

Determining the weights and biases of the MLP is formulated as an optimization problem with the objective of minimizing an aggregation of the individual losses $\ell(\theta; \mathbf{x}, \mathbf{y})$ in the training set. The loss function in regression problems is usually chosen to be Mean Squared Error (MSE). Gradient-based methods like stochastic gradient descent and its variants may be used to solve the optimization problem where the back propagation method can be used to find the gradients of the neural network [12]. Dropout is a popular regularization technique for neural networks to limit over-fitting. For every weight update, hidden units in a layer have a chance (determined by the dropout rate) to be ignored during that cycle.

B. Shapley values

Shapley values are a concept from cooperative game theory where players (features) work together to form some pay-out (prediction). The Shapley value of the j th feature in the context of a model f and an instance $\mathbf{x}$ gives the weighted average of all possible contributions the feature may have to the model output. For all possible coalitions, the prediction of the model is found *with* and *without* the presence of the feature in question. The exact Shapley value is defined by:

$$\phi_j = \sum_{S \subseteq F \setminus \{j\}} \frac{|S|!(|F| - |S| - 1)!}{|F|!} [f_{S_j}(\mathbf{x}_{S_j}) - f_S(\mathbf{x}_S)], \quad (4)$$

where S is a subset of the set of features F without j , and $S_j = S \cup \{j\}$ is the subset with j added. Likewise, f_{S_j} is a model built with features in S_j as inputs, and f_S is a model built with S features. This implies that for every subset, the model has to be retrained twice; once for the subset containing j and once without. This is computationally expensive since there are $2^{|F|}$ different coalitions to evaluate.

SHAP introduced by [13] proposed the so-called SHAP values which are the Shapley values of a conditional expectation function of the original model. Instead of retraining the model for every subset, SHAP values approximate the model's output for a subset of inputs with the expected value of the original model conditioned on that subset. I.e., define that $f_{\mathbf{x}}(S) = f(h_{\mathbf{x}}(\mathbf{z}')) = E[f(\mathbf{x})|\mathbf{x}_S]$, where $h_{\mathbf{x}}$ is the input mapping function and $\mathbf{z}'$ is a binary vector representing whether a feature is missing or present (0 or 1). SHAP values satisfy three important properties of feature importance that other methods do not such as local accuracy, missingness and consistency [13].

III. METHODOLOGY

AI models function as black box models for most researchers and professionals, who use them in domains like energy, healthcare and robotics. Most AI methods lack inherent details and explanations which can help domain experts make better decisions and interpretations. I.e., they lack the ability to keep human agents "in the loop". XAI provides an opportunity to transfer the AI-based black box models to more explainable "gray box" models. Fig. 1 shows a top-level flow chart of the methodology using an XAI tool for a load forecasting model.

A. Multilayer perceptron forecasting model

Day-ahead load forecasting is a multi-step time series forecasting problem. A Multi-Input Multi-Output (MIMO) forecasting strategy as described in [14] was chosen as the forecasting strategy. The advantage of this approach is that it only requires one predictor for all twenty-four output values, thus mitigating the amount of computing power needed compared to training one predictor for each hour. The disadvantage is that the complexity of each hour has to be represented by a single model, which may be limited in capacity. The input features to the model are the 24 previous (historical) load values prior to the forecast; 24 temperature predictions in

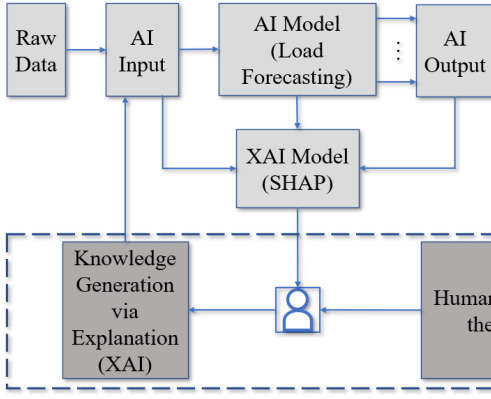


Fig. 1: Explainable artificial intelligence flow

central Oslo for each output hour; 24 temperatures in Lillehammer for each output hour; and encoded day of the week features (i.e., d). The outputs of the model are the 24 load values one day ahead. See Figure 2 for a timeline of the various quantities.

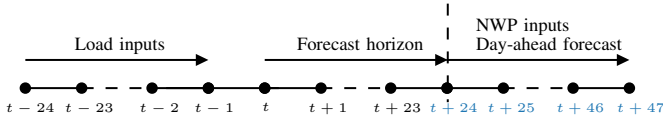


Fig. 2: Timeline of the inputs and outputs of the day-ahead forecasting model. The termin time is denoted as t and the lead time is twenty-four hours (day-ahead).

The forecasting strategy was defined so that a day-ahead forecast is made once each day at 12PM (henceforth referred to as the termin time). Let t denote the termin time and $P^{(t)}$ the average load of that hour, then the MIMO model may be described by:

$$\hat{P} = f(P, T_o, T_l, d), \quad (5)$$

where $f : \mathbb{R}^{79} \rightarrow \mathbb{R}^{24}$; $\hat{P}$ is the vector of predicted values, i.e., $(\hat{P}^{(t+24)}, \hat{P}^{(t+25)}, \dots, \hat{P}^{(t+47)})$; P is the vector of historical load values, $(P^{(t-24)}, P^{(t-23)}, \dots, P^{(t-1)})$; T_o and T_l are the vectors of forecast temperatures in Oslo and Lillehammer, respectively; and d is the vector of one-hot encoded day of the week variables, i.e., $d_0, d_1, \dots, d_6$ (Monday-Sunday).

Equation (5) was implemented in Keras as an MLP with two hidden layers and dropout. For neural network hyper-optimization, performing a random search is recommended as opposed to doing a grid search over the hyper-parameters [15]. Thus, an initial random search was made to determine optimizer, activation functions, dropout and number of units. From the first random search results, the search was narrowed down and another one was performed based on the first results. The final optimizer was Nadam; the hidden layer activation functions were eLU and tanh respectively and the number of units were set to 384 and 456.

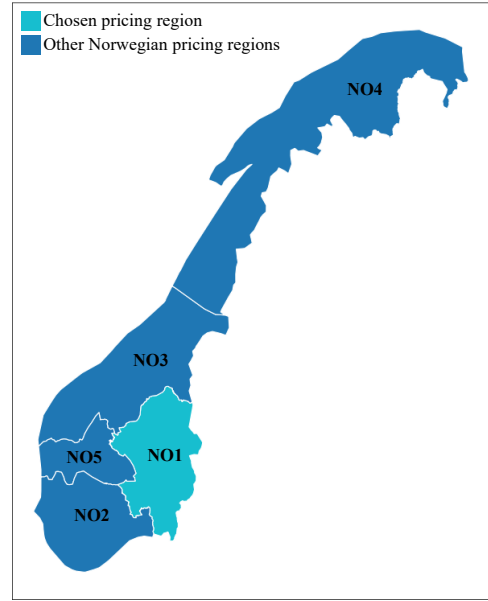


Fig. 3: The Norwegian pricing regions.

As training data, historical data from 2014-2019 was used, where 2019 was chosen for as validation data for early-stopping during training and for hyper-optimization scores. Lastly, 2020 was used as the hold-out (test) set in order to prevent any risk of look-ahead bias. The model was only trained once, as opposed to updating the model with newly observed samples when evaluating the test year. By choosing this approach, the model does not get access to new information contained in the test year as time progresses, but in return only has to be trained one time. Moreover, the year of 2020 has some interesting changes in consumption patterns due to the Corona virus which the model will not learn.

B. Data collection and pre-processing

Historical Numerical Weather Predictions (NWP) are publicly available from the Norwegian Meteorological Institute (MET) in NetCDF4 format [16]. Historical temperature predictions in Oslo and Lillehammer were extracted from MET's THREDDS data server from 2014 to 2020. The newest NWP that would have been available at a historical termin time was chosen in order to reflect a real forecasting scenario.

The market data for the NO1 pricing region (Figure 3) was downloaded from Nord Pool [17]. The platform offers hourly electrical consumption values for every pricing region within Norway. The consumption is found from the sum of production and flow into the given area, where each data point is the total consumption for the upcoming hour in MWh. Hence, also the average electrical load of that hour in MW.

Historical electric load and temperature data were reshaped into daily samples of 24 features, one for each hour of the day (see Figure 2) and were then standardized to fit the training data.

C. Improving Easter week using SHAP

Easter is celebrated as a public holiday where electric consumption patterns diverge from typical patterns. People wake up later, hence the consumption is lower and shifted to later in the day. If the model is not taught how to distinguish public holidays from any other day, the predictions for those days will most likely be poorer. Thus, another one-hot encoded variable indicating whether the day-ahead is within Easter was added to the model. I.e., (5) was extended such that

$$\hat{P} = f(P, T_o, T_l, d, e), \quad (6)$$

where e is the Easter variable. It was defined such that $e = 1$ if the day-ahead falls within Palm Sunday up to second Easter day, and $e = 0$ if not. The hyper-parameters were kept the same for the second model.

Kernel SHAP was used in order to see how the model is affected by the added feature and to confirm if it was using it the way it was intended to. Kernel SHAP defines the absence of a feature by replacing it with a randomly chosen value from a background set, which is supposed to represent the distribution of the features. It then estimates the SHAP values by a chosen number of samples (how many different subsets to evaluate) by fitting the explained instance to a linear regression model.

To interpret the effect of adding the Easter feature, SHAP summary plots (see Figure 5 and Figure 6) were made for the thirteenth output hour of the forecasting model (i.e., 12AM of the target day) for all the nine days defined as Easter. Two different models were trained: one with the variable, and one without. The ten most influencing features were shown, where each SHAP value denotes how much the feature contributed to the prediction compared to the average prediction of the background set. Note however, that since the zero-vector was passed as the background set, the contribution of each feature is compared against the zero-vector prediction, *not* the average prediction of some real data set.

IV. RESULTS AND DISCUSSION

For the evaluation of the results, Mean Absolute Percentage Error (MAPE) and Root Mean Square Error (RMSE) parameters were chosen. Table I shows the results for the validation Easter (2019) and test Easter (2020), for the models without and with the Easter variable. It also shows the metrics for the entire test year for both models. There is a clear improvement in both MAPE and RMSE values for both models. However, it is surprising that Easter 2020 has the greatest increase in performance (31.7%) compared to Easter 2019 (19.3%). One would expect that Easter 2019 would have the greatest increase in performance since it was used as validation data. This may be attributed to the unexpected rapid decrease in consumption as seen in Figure 4. It may also be argued that the improvement for the entire test set may be due to chance due to the stochastic nature of neural network optimization. The Easter improvements however are greater than what is usually attributed to normal variation in performance.

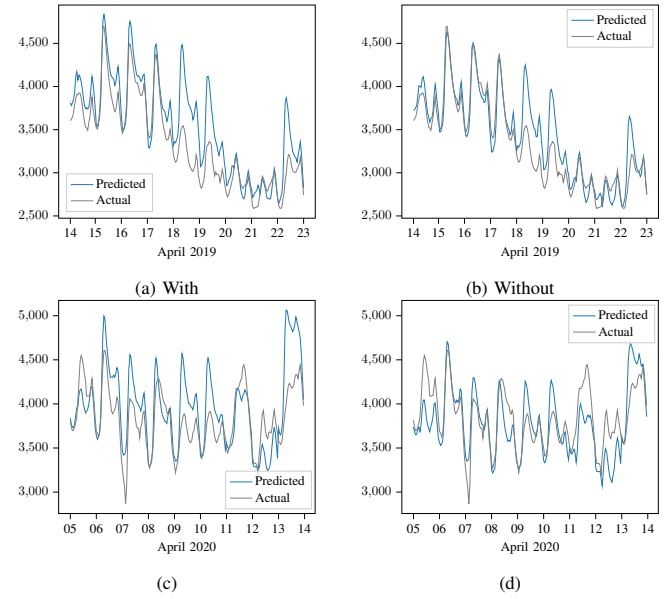


Fig. 4: The time series of 2019 and 2020 Easter weeks before (left) and after (right) adding a one-hot encoded variable to the day-ahead forecasting model. The time series show average hourly active load, P (MW); forecast values are in blue, while actual observations are in gray.

The time series of Easter and the corresponding SHAP summary plots for 2019 and 2020 are shown in Figure 4. There is a noticeable reduction in deviation from the actual values when adding the Easter variable both for the validation year, and the test year. From visual inspection it can be observed that the addition of the feature leads to slightly reduced predictions over the entire Easter. This observation is confirmed by considering the SHAP summary plots made for both years; the Easter variable is assigned a relatively large negative SHAP value and the greatest importance for both years. It is however difficult to come to conclusions regarding the importance of the other features. Outside of the obvious contribution of day of the week variables, such as lower consumption on Sundays (negative SHAP value) and higher on Mondays (positive SHAP value), it is hard to interpret the contribution of the past load and temperature features which requires more future work.

Lastly, it is important to note that adding the Easter variable as an input to the forecasting model was *expected* to increase the forecasting accuracy. Hence, it is not surprising that Kernel SHAP attributes the Easter variable a large importance. It does however confirm the fact that the model is using the variable the way it was supposed. SHAP serves as a useful tool for model improving, debugging, feature engineering and interpretation.

TABLE I

Metrics for the validation (2019) and test (2020) years.

	Easter 2019		Easter 2020		Entire 2020	
	MAPE	RMSE	MAPE	RMSE	MAPE	RMSE
Without	6.43	320	7.19	312	3.93	193
With	5.13	258	4.92	224	3.71	184

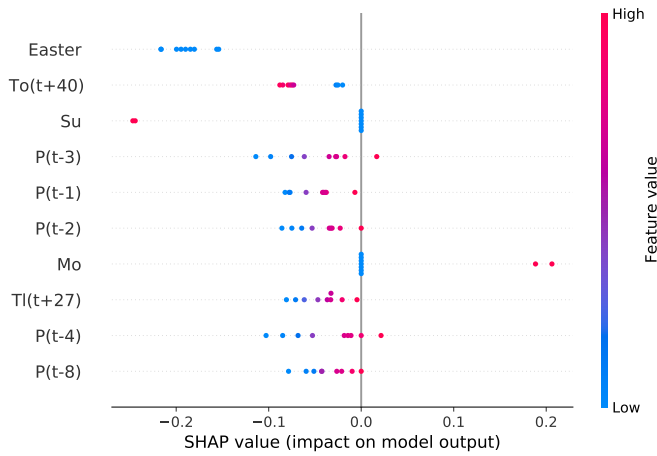


Fig. 5: SHAP summary plot for 2019 Easter. Each dot represents the SHAP value pertaining to the thirteenth output hour (12AM) of a day in Easter week.

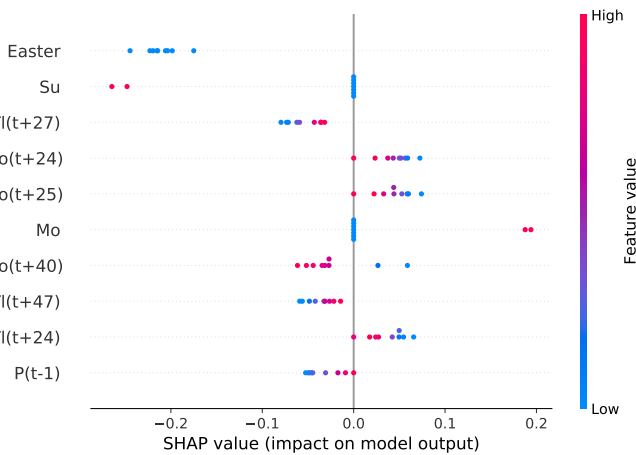


Fig. 6: SHAP summary plot for 2020 Easter. Each dot represents the SHAP value pertaining to the thirteenth output hour (12AM) of a day in Easter week.

V. CONCLUSION

Recent changes in energy sector which is mainly fuelled by required policy changes across many governments for achieving low carbon emissions and contributing towards various United Nations Sustainable Development Goals (UN-SDG) such as affordable clean energy (Goal 7) and sustainable cities (Goal 11) resulted in high integration of volatile DER. Due to volatility and distributed nature of these energy resources, digitalization trends such as AI for forecasting both electricity generation and load side became more important. Beside generation side uncertainties due to fluctuated behaviour of renewables, high penetration of electric vehicles (EV) lead some additional uncertainty with considerable impact in the consumption side.

With the increasing trend towards digitalization, AI has penetrated various industries such as energy, health and robotics. However, recent advances in the field of AI such as Deep Neural Networks (DNN) resulted in more black-box type models to be utilized. Such AI models lack the details and explanations

which can potentially help domain experts to make better decisions and interpretations by keeping the human agents in the loop.

In this study, the use of SHAP as an XAI tool for a functional day-ahead load forecasting is presented. The selected demonstration region was Norwegian Price Region 1 (NO1). The contribution of the Easter variable on forecast accuracy was shown and described by the use of one of SHAP's visualization tools. Detailed investigations on the other feature of importance will be carried out in the future studies. More work is also needed to test out other SHAP calculation methods.

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